

1.2.3 Units

☐ The units of data storage:

- Bit
- Nibble (4 bits)
- Byte (8 bits)
- Kilobyte (1,000 bytes or 1 KB)
- Megabyte (1,000 KB)
- Gigabyte (1,000 MB)
- Terabyte (1,000 GB)
- Petabyte (1,000 TB)

☐ How data needs to be converted into a binary format to be processed by a computer

☐ Data capacity and calculation of data capacity requirements

Required

- ✓ Why data must be stored in binary format
- ✓ Familiarity with data units and moving between each
- ✓ Data storage devices have different fixed capacities
- ✓ Calculate required storage capacity for a given set of files
- ✓ Calculate file sizes of sound, images and text files
 - sound file size = sample rate x duration (s) x bit depth
 - image file size = colour depth x image height (px) x image width (px)
 - text file size = bits per character x number of characters

Alternatives

- Use of 1,024 for conversions and calculations would be acceptable
- Allowance for metadata in calculations may be used